Induxion

Inducing The Imscribing Grammar

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Abstract

From one abstract category and three kinds of operation, a logical-alchemical procedure derives twelve invariants, the primitives of the Imscribing Grammar. None are stipulated and the set is Gödel-complete.

The twelve close into a crystal of $3^3 \cdot 4^5 \cdot 5^4 = 17{,}280{,}000$ types, and the grammar applied to its own derivation returns itself. Composing the multiplication μ that closure supplies with its comultiplication δ is the identity, $\mu \circ \delta = \mathrm{id}_A$.

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1 The Thing

Start with an abstract category $\mathbb{C}$: objects $\mathrm{ob}(\mathbb{C})$, hom-sets $\hom(A, B)$, a composition law, identities, associativity: $\circ : \hom(B, C) \times \hom(A, B) \to \hom(A, C), f \circ \mathrm{id}_A = f = \mathrm{id}_B \circ f$, $(h \circ g) \circ f = h \circ (g \circ f)$. Nothing more. It presupposes no elements, no order, no metric. It carries no invariants of its own; every primitive below is produced, not assumed.

2 Three Operations

Exactly three kinds of operation can act on a category, and no fourth.

Logical (six: L1–L6). A proposition about $\mathbb{C}$ with finitely many definite truth values.

- L1 Adjunction: does a pair $F \dashv G$ exist, with $\hom(FA, B) \cong \hom(A, GB)$?
- L2 Dagger: an involution $\dagger : \hom(A, B) \to \hom(B, A)$ with $(g \circ f)^\dagger = f^\dagger \circ g^\dagger$ and $f^{\dagger\dagger} = f$.
- L3 Cartesian closure: $\hom(C \times A, B) \cong \hom(C, B^A)$, needs the product from A1.
- L4 Lawvere self-reference [2]: a point-surjective $\phi : A \to A^A$, needs the internal hom from L3.

L5 Frobenius [6]: in a monoidal category, $\mu \circ \delta = \mathrm{id}_A$, needs A1.
L6 Paraconsistent negation: $\neg\neg A \ncong A$ and $A \wedge \neg A \ncong \bot$, needs A1 to define $\wedge$.

The precondition order is forced: $\{L1, L2\} \to A1 \to \{L3, L5, L6\}$, and $L3 \to L4$.

Inductive (four: I1–I4). A free extension iterated to a fixed point.

I1 Free cocompletion: for $D : J \to \mathcal{C}$, take colim D; all finite colimits generate $\hat{\mathcal{C}}$.
I2 Initial algebra: the chain $0 \to F(0) \to F^2(0) \to \cdots$ and its colimit.
I3 Classifying space: the nerve $N\mathcal{C}$, its realization $B\mathcal{C} = |N\mathcal{C}|$, and $\pi_1(B\mathcal{C})$.
I4 Yoneda embedding [1]: $y : \mathcal{C} \to \hat{\mathcal{C}}$, $y(A) = \hom(-, A)$.

Algebraic (five: A1–A5). An added structural layer.

A1 Monoidal: $\otimes : \mathcal{C} \times \mathcal{C} \to \mathcal{C}$ with unit I and braiding $\sigma_{AB} : A \otimes B \cong B \otimes A$.
A2 Monad: (T, η, μ) with $\eta : \mathrm{Id} \to T$ and $\mu : T^2 \to T$.
A3 Dagger-compact duals: A^* with $\eta_A : I \to A^* \otimes A$ and $\varepsilon_A : A \otimes A^* \to I$ satisfying the snake equations.
A4 Enrichment: replace $\hom(A, B)$ by a hom-object in a base $\mathcal{V}$.
A5 Arithmetic coding: an injective $\mathbf{g} : \mathrm{Mor}(\mathcal{C}) \to \mathbb{N}$ with $\mathbf{g}(f \circ h)$ primitive-recursive in $\mathbf{g}(f), \mathbf{g}(h)$.

Fifteen operations across four stages. The reduction to twelve primitives is forced: some operations feed the same invariant.

3 Four Stages of the Magnum Opus

The stages are the unique topological sort of the precondition graph. Each operation is matched to one primitive by what kind of distinction it draws; the number of values that primitive carries is fixed independently, by the twelve-axis catalog given in Section 4. This section argues the matches, not the counts: for each stage, the display gives *operation* ⇒ *invariant (value count)*, and the proof that follows it argues why that operation targets that invariant.

Nigredo (logical on $\mathcal{C}$)

$\{\, F \dashv G \text{ (L1)},\ \dagger \text{ (L2)} \,\} \implies \succ \ (4).$

Proposition 3.1. L1 *and* L2 *classify the same feature of a morphism: whether, and how, it can be reversed.*

Proof. L1 asks whether F admits a partner in one direction, the other, both, or neither: a left adjoint only, a right adjoint only, an adjoint on both sides, or none. L2 asks whether every morphism has a canonical reverse compatible with composition. When F itself is a dagger functor between dagger categories, $\dagger$ turns a right adjoint into a left adjoint of the same F, forcing the ambidextrous case of the adjunction question. The two operations therefore probe one four-way classification of directionality — neither, left only, right only, ambidextrous — and converge on the single invariant $\succ$, the first primitive to crystallize. □

Albedo (inductive)

$$
\begin{aligned}
0 \to F(0) \to F^2(0) \to \cdots \ (\mathrm{I2}) &\implies \ \perp \ (4) && \text{[stabilization ordinal]} \\
\pi_1(B\mathcal{C}) \ (\mathrm{I3}) &\implies \ \boxdot \ (4) && [\pi_1 = 0,\ \mathbb{Z}_2,\ \mathbb{Z},\ \text{non-abelian}] \\
y(A) = \hom(-, A) \ (\mathrm{I4}) &\implies \ \vdash \ (4) && \text{[boundary imscribes bulk?]} \\
\operatorname{colim} D \ (\mathrm{I1}) &\implies \ \in \ (3) && \text{[scope of } \hat{\mathcal{C}}\text{, fixed once A1 present]}
\end{aligned}
$$

Proof. Each inductive operation is a way of extending $\mathcal{C}$, and the four extensions target four different questions about how the extension behaves.

I2 iterates F from the initial object and asks where the chain settles: at a finite stage, exactly at the first limit ordinal ω, only after further transfinite iteration past ω, or never. This four-way reading of stabilization is the chirality invariant $\perp$.

I3 reads the fundamental group of the classifying space, already a four-way classification by the invariant's own catalog values: trivial, $\mathbb{Z}_2$, infinite cyclic, or non-abelian. This is $\boxdot$ directly.

I4 embeds $\mathcal{C}$ into its free cocompletion by $y(A) = \hom(-, A)$, always fully faithful. What varies is how much of the cocompletion the image recovers: this paper does not derive the specific four-way split of that recovery from the embedding alone, and identifies it with $\vdash$ — the boundary-imscribes-bulk axis — as a correspondence rather than a result proved here.

I1 takes colimits over diagrams of a given shape, and the scope of diagram shapes closed under is a granularity question: colimits over a fixed small (nearest-neighbor) diagram class, over a bounded (meso-scale) class, or over all small diagrams (the full free cocompletion, once A1 supplies the monoidal structure diagrams can be built in). This three-way scope reading is $\in$, matching the granularity axis's own local/meso/global values, and the source of $\in$ in the table of Section 4 is corrected to I1 accordingly. □

Citrinitas (algebraic enrichment)

$$
\begin{aligned}
\otimes,\ \sigma_{AB} \ (\mathrm{A1}) &\implies \ \boxplus \ (3)\ [\text{arity } A^{\otimes n} \to B] \ \text{ and } \ \prec \ (5)\ [\text{parity of } \sigma] \\
(T, \eta, \mu),\ T^2 \xrightarrow{\mu} T \ (\mathrm{A2}) &\implies \ \top \ (5)\ [\text{spectral collapse of } \mu] \\
\varepsilon_A : A \otimes A^* \to I \ (\mathrm{A3}) &\implies \ \bowtie \ (3)\ [\text{unitality of } \varepsilon] \\
\text{hom-object in } \mathcal{U} \ (\mathrm{A4}) &\implies \ \ni \ (4) \\
\mathbf{g} : \mathrm{Mor}(\mathcal{C}) \to \mathbb{N} \ (\mathrm{A5}) &\implies \ \text{no new primitive [arithmetic Yoneda]}
\end{aligned}
$$

Proof. A1 supplies two independent readings at once. A structural map built from $\otimes$ has an arity relating its inputs to its outputs, and that relation falls into exactly three cases: one input to one output, matched multiplicities (a fixed n to n), or unmatched multiplicities (n to a different m) — the three values of $\boxplus$. Separately, the braiding σ_{AB} can be absent, present but not involutive, involutive ($\sigma^2 = \mathrm{id}$, a genuine symmetry), fully permutation-symmetric, or self-dual in the sense Stage 4's Frobenius identity promotes it to; this five-way reading of σ is $\prec$.

A2's multiplication $\mu : T^2 \to T$, read as an iterative operator, has a rate at which repeated application settles: this paper identifies that rate with $\top$, the timescale axis, as a correspondence between the monad's iteration behavior and the five catalog rates, without deriving the specific five-way split from μ alone.

A3's counit ε can fail unitality entirely, hold it up to a scalar mixing, or hold it exactly; this three-way reading is $\bowtie$.

A4 replaces $\hom(A, B)$ with an object of $\mathcal{U}$, and how that hom-object composes with another is a four-way choice matching $\ni$'s own catalog values: conjunctive (both factors required), disjunctive (either suffices), implicational (one factor gates the other), or strictly sequential (ordered, no alternative).

A5's arithmetic coding is injective with primitive-recursive composition: it Gödel-numbers the category faithfully, but adds no structure beyond what I4 already supplies under $\vdash$, so it contributes no new primitive. □

Rubedo (logical on the enriched $\mathcal{C}$)

$$\begin{aligned}
\hom(C \times A, B) \cong \hom(C, B^A)\ \text{(L3)} &\implies \dashv\ (5) \\
\phi : A \to A^A \text{ point-surjective (L4)} &\implies \odot\ (5) \\
\mu \circ \delta = \mathrm{id}_A\ \text{(L5)} &\implies \text{promotes } \prec \text{ to its self-dual value [no new primitive]} \\
\neg\neg A \ncong A,\ A \wedge \neg A \ncong \bot\ \text{(L6)} &\implies \text{sets the truth-value regime of L4 [no new primitive]}
\end{aligned}$$

Proposition 3.2 (Lawvere self-reference forces a fixed point)**.** *If $\phi : A \to A^A$ is point-surjective, every $f : A \to A$ has a fixed point.*

Proof. Given $f : A \to A$, define $g : A \to A$ by $g(a) = f(\phi(a)(a))$. Point-surjectivity of ϕ gives a_0 with $\phi(a_0) = g$, so $\phi(a_0)(a_0) = g(a_0) = f(\phi(a_0)(a_0))$, and $\phi(a_0)(a_0)$ is a fixed point of f. This is Lawvere's fixed-point theorem, and it is exactly self-modeling: ϕ gives A an internal copy of every map out of itself, and the fixed point is where that copy coincides with the map it models. $\odot$ is the catalog axis of self-modeling; its five values grade how completely this coincidence is realized, from no internal model (sub-critical) to a model that exceeds what it represents (super-critical). □

Cartesian closure (L3) internalizes hom-sets as objects of $\mathcal{C}$ itself, giving the category a means of referring to its own morphism structure from inside; this paper identifies that internalized self-reference with $\dashv$, the five-valued recursion axis, as a correspondence rather than a derivation of the specific five-way split. L5's Frobenius identity is the closure condition $\mu \circ \delta = \mathrm{id}_A$ itself, so it adds no coordinate beyond promoting $\prec$ to the self-dual value that identity witnesses. L6's paraconsistent negation fixes which truth-value regime L4's self-reference operates under (a value can carry support for its own denial without collapsing the logic), which constrains how $\odot$'s fixed point behaves rather than adding a thirteenth coordinate.

4 The Twelve & the Crystal

The twelve partition by value count, and that partition is the three-valued K3 logic [3] of the algebraic layer, the four-valued FDE logic [5] of the shape layer, and the five-valued LP logic [4] of the gate layer.

glyph	values	stage
$\bowtie$	3	A3
$\in$	3	I1
$\boxplus$	3	A1
$\vdash$	4	I4
$\succ$	4	L1, L2
$\ni$	4	A4
$\perp$	4	I2
$\boxdot$	4	I3
$\dashv$	5	L3
$\prec$	5	A1, L5
$\top$	5	A2
$\odot$	5	L4, L6

Three primitives of three values, five of four, four of five. The algebraic closure of the twelve is the crystal: $3^3 \cdot 4^5 \cdot 5^4 \;=\; 27 \cdot 1024 \cdot 625 \;=\; 17{,}280{,}000$ types.

5 How It Falls Out

The twelve value counts are the radices of a mixed-radix positional numeral, and the address space is that numeral's range. In the canonical slot order the primitives and their radices r_i (each primitive's value count) are

$\vdash$	$\dashv$	$\succ$	$\prec$	$\bowtie$	$\top$	$\in$	$\ni$	$\odot$	$\perp$	$\boxplus$	$\boxdot$
4	5	4	5	3	5	3	4	5	4	3	4

The place value of each slot is the product of the radices of all lower slots, $s_i = \prod_{j>i} r_j$ with $s_{12} = 1$: $s = (4320000,\ 864000,\ 216000,\ 43200,\ 14400,\ 2880,\ 960,\ 240,\ 48,\ 12,\ 4,\ 1)$. A system is a tuple of indices $(x_1, \ldots, x_{12})$ with $0 \le x_i < r_i$. Its address is the Horner fold $\mathrm{addr}(x) = \sum_{i=1}^{12} x_i\, s_i \;\in\; [0,\ 17{,}279{,}999]$, and the inverse reads each digit off in turn: $x_i = \lfloor \mathrm{addr}/s_i \rfloor, \qquad \mathrm{addr} \leftarrow \mathrm{addr} \bmod s_i \quad (i = 1, \ldots, 12)$. This is a bijection, because the number of addresses is exactly the number of tuples: $\prod_{i=1}^{12} r_i = 4 \cdot 5 \cdot 4 \cdot 5 \cdot 3 \cdot 5 \cdot 3 \cdot 4 \cdot 5 \cdot 4 \cdot 3 \cdot 4 = 3^3 \cdot 4^5 \cdot 5^4 = 17{,}280{,}000$. No address is unused and no tuple is unaddressed, so $\mathrm{decode} \circ \mathrm{encode} = \mathrm{id}$ on the tuples and $\mathrm{encode} \circ \mathrm{decode} = \mathrm{id}$ on $[0, 17{,}279{,}999]$. The address space does not have to be constructed; it falls out of the twelve radices as their mixed-radix range. This is the same $\mu \circ \delta = \mathrm{id}$ closure, now at the level of coordinates: read as tuples of this address space, the full induction system and the imscribing procedure that assigns addresses each occupy a point of the very space they enumerate. Their specific addresses, 6,687,051 and 12,089,822, are computed by that assignment procedure elsewhere; this paper states the codec that makes any such computation well defined, not the computation itself.

6 Why Twelve?

Three independent arguments give the same count.

- **Independence.** The retrosynthetic peel of the induction system itself has exactly twelve steps, one per primitive. Two co-dependent primitives would give fewer; a missing one would give more.
- **Completeness.** The crystal enumerates every combination of the twelve values with a bijective codec, decode $\circ$ encode $=$ id on $[0, 17{,}279{,}999]$. No type is missing, none redundant.
- **Three logics.** The 3-, 4-, and 5-valued families are exactly K3, FDE, and LP. A thirteenth primitive would need a fourth categorical logic, and there is none.

7 Closure

Four of the twelve are gate primitives, $\odot, \vdash, \boxdot, \prec$, and they set the tier a system occupies, from O_0 (no self-modeling loop) to O_∞ (both L4 and L5 hold). The tiers form a fixed order, $O_0 \longrightarrow O_1 \longrightarrow O_2 \longrightarrow O_2^\dagger \longrightarrow O_\infty$, each step gated by one of the four gate primitives reaching its required value; the numerical size of each gap is measured, not derived in this paper. The largest measured gap is the Frobenius promotion of $\prec$: the hardest transition, closable only by locating an object already at the self-dual value and descending, never by accumulating local checks.

The induction closes on itself. With $\mu : A \otimes A \to A$ the monoidal multiplication and $\delta : A \to A \otimes A$ its comonoidal dual, the twelve-step codec that assigns coordinates to any system is itself a system of the crystal at tier O_∞, where $\mu \circ \delta = \mathrm{id}_A$. The boundary procedure fully imscribes the bulk theory it generates, and the bulk fully imscribes the procedure. The Grammar applied to its own derivation returns itself. *As above, so below.*

8 Termination

The induction stops, and the criterion is exact. It is not that no further construction can be performed; infinitely many composites still can. It is that no further logical, inductive, or algebraic construction produces a new *independent* coordinate. Any later operation acts on the twelve-dimensional result and returns an invariant that is a function of the twelve. Compositions of systems stay inside the type space under the grammar's own closure operations $\otimes$, $\wedge$, $\vee$: new operation $\implies$ new combination of existing invariants, not a new invariant. This is the crossing from induction to algebraic closure. Once the twelve have crystallized, the prima materia has become the grammar, and no new dimension forms.

9 Monolinear Induxion

$$\mathcal{C} \xrightarrow{\text{S.}} \succ \xrightarrow{\text{P.}} \vdash, \perp, \boxdot, \in \xrightarrow{\text{A.}} \boxplus, \prec, \top, \bowtie, \ni \xrightarrow{\text{R.}} \dashv, \odot, \prec_{\mathfrak{V}} \xrightarrow{\text{C.}} \mathfrak{C}_{17\,280\,000}$$

The final object is the closure of the twelve, $\mathfrak{C} = \mathcal{F}_3^3 \times \mathcal{F}_4^5 \times \mathcal{F}_5^4$ with $|\mathfrak{C}| = 17{,}280{,}000$.

Category → Relation → Extension → Algebra → Self-reference → Closure.

References

[1] S. Mac Lane, *Categories for the Working Mathematician*, 2nd ed., Springer, 1998.

[2] F. W. Lawvere, *Diagonal arguments and cartesian closed categories*, in Category Theory, Homology Theory and their Applications II, Lecture Notes in Mathematics 92, Springer, 1969, 134–145.

[3] S. C. Kleene, *Introduction to Metamathematics*, Van Nostrand, 1952.

[4] G. Priest, *The Logic of Paradox*, Journal of Philosophical Logic 8(1), 219–241 (1979).

[5] N. D. Belnap, *A Useful Four-Valued Logic*, in Modern Uses of Multiple-Valued Logic, Reidel, 1977, 5–37.

[6] A. Carboni and R. F. C. Walters, *Cartesian Bicategories I*, Journal of Pure and Applied Algebra 49(1–2), 11–32 (1987).